

Yuankai (William) He

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RESEARCH INTERESTS

Cyber-Physical Intelligent Systems; V2X; ITS; Robotics; Digital Twin.

EDUCATION

- **University of Delaware**, Ph.D. in Computer Science. Advisor: Weisong Shi. Expected Dec. 2026
- **Wayne State University**, M.S. in Robotics. Advisor: Weisong Shi. May 2023
- **Boston University**, B.S. in Computer Engineering. May 2019

HONORS & RECOGNITION

- **Frank A. Pehrson Graduate Student Award for Outstanding Computer Science Research.**
UD Computer & Information Sciences' most prestigious research award; single recipient. 2026
- **Daniel L. Chester Graduate Fellowship Award.**
University of Delaware. Consecutive-year recipient. 2025, 2026

RESEARCH RECORD & LEADERSHIP

- **Robust physical intelligence from onboard execution to physical validation.**
Connects timely onboard execution, shared sensing and computing, and repeatable sim-to-real evaluation on two full-scale autonomous vehicles, C-V2X/private 5G, live roadway feeds, and digital twins.
- **Multiple first-author papers and invited research talks.**
Published first-author work at *IEEE IV*, *IEEE T-VT*, and *IEEE Computer*, with additional papers at *ICDCS*, *IEEE MOST*, and *HotStorage*; invited to present at *IEEE IV*, *UCI*, and *William & Mary*.
- **Graduate proposal contributor on six funded proposals totaling \$1,140,416.**
Lead graduate researcher on five associated projects spanning NSF, TIER IV, and Toyota.
- **Springer textbook and graduate autonomous-driving curriculum.**
Co-authored *Introduction to Autonomous Driving*; designed and taught UD's graduate autonomous-driving course; and developed and taught the Autonomous Driving Academy.

PUBLICATIONS

Citation counts: Google Scholar, September 29, 2026.

Book

- W. Shi, **Y. He**. *Introduction to Autonomous Driving*. Springer, 2025. [DOI](#) [Demo](#) [Book Companion](#)

Peer-Reviewed Journal Articles

- **Y. He**, B. Wu, Z. Dong, J. Wan, W. Shi. "Towards C-V2X Enabled Collaborative Autonomous Driving." *IEEE Transactions on Vehicular Technology*, 72(12):15450–15462, 2023. Cited by: **83** [DOI](#) [PDF](#)
- S. Lu, **Y. He**, W. Shi. "Vehicle Computing: An Emerging Computing Paradigm for the Autonomous Driving Era." *Journal of Computer Research and Development*, 62(1):2–21, 2025. Cited by: **2** [DOI](#) [PDF](#)
- **Y. He**, W. Shi. "A Vision for Transformative Intersections." *IEEE Computer*, 57(12):58–68, 2024. Cited by: **2** [DOI](#) [PDF](#)

Peer-Reviewed Conference and Workshop Papers

- Z. Tian, **Y. He**, B. Tian, R. Zhong, E. Foorginejad, W. Shi. "ICAT: An Indoor Connected and Autonomous Testbed for Vehicle Computing." *IEEE International Conference on Mobility, Operations, Services and Technologies (MOST)*, pp. 242–250, 2024. Cited by: **19** [DOI](#) [PDF](#)
- Y. Wang, **Y. He**, R. Wang, W. Shi. "Quantitative Analysis of Storage Requirement for Autonomous

- Vehicles.” *Proceedings of the 16th ACM Workshop on Hot Topics in Storage and File Systems (HotStorage '24)* (ACM SIGOPS workshop), pp. 71–78, 2024. Cited by: **12** [DOI](#) [PDF](#)
- B. Wu*, **Y. He***, Z. Dong, J. Wan, J. Zhang, W. Shi. “To Turn or Not To Turn, SafeCross is the Answer.” *IEEE 42nd International Conference on Distributed Computing Systems (ICDCS)* (IEEE Computer Society flagship conference), pp. 414–424, 2022. (*Equal contribution) Cited by: **8** [DOI](#) [PDF](#)
 - **Y. He**, W. Shi. “A Faster and More Reliable Middleware for Autonomous Driving Systems.” *IEEE Intelligent Vehicles Symposium (IV)* (IEEE ITSS flagship conference), 2026. Cited by: **7** [PDF](#)
 - **Y. He**, H. Chen, I. Safro, W. Shi. “A Graph-Based Metric on Dataset Diversity and Redundancy Evaluation.” *IEEE Intelligent Vehicles Symposium (IV)* (IEEE ITSS flagship conference), 2026. Cited by: **0** [PDF](#)
 - **Y. He**, Y. Wang, B. Tian, W. Shi. “DaVoS: An Instantaneous Safety Model for Autonomous Vehicles.” *Physical Intelligence: Systems and Applications (PISA) Workshop at IEEE MOST*, pp. 256–262, 2026. Cited by: **0** [DOI](#) [PDF](#)
 - J. Shi, **Y. He**, Y. Wang, W. Shi. “Can I Park Here? Understanding Complicated Parking Signs on the Street.” *IEEE MOST*, pp. 107–117, 2026. Cited by: **0** [DOI](#) [PDF](#)

Manuscript Under Review

- **Y. He**, H. Chen, W. Shi. “An Advanced Framework for Ultra-Realistic Simulation and Digital Twinning for Autonomous Vehicles.” Cited by: **11** [PDF](#)
- **Y. He**, W. Shi. “CARScenes: Semantic VLM Dataset for Safe Autonomous Driving.” Cited by: **2** [PDF](#)
- Y. Wang, **Y. He**, B. Tian, L. Xian, W. Shi. “DAVOS: An Autonomous Vehicle Operating System in the Vehicle Computing Era.” Cited by: **0** [PDF](#)
- Y. Wang, **Y. He**, W. Shi. “AVS: A Computational and Hierarchical Storage System for Autonomous Vehicles.” Cited by: **0** [PDF](#)

PROPOSAL WRITING EXPERIENCE

Graduate proposal contributor on three funded NSF proposals (\$970,416) and three industry proposals (\$170,000).

NSF Projects

- **NSF CNS Collaborative Research: CSR: Small; No. 2426482.**
CADaaS: Deterministic Communication and Predictive Computation for Connected Autonomous Driving as a Service.
Support: \$273,555. Mar. 2025 – Feb. 2028
Role: Lead Graduate Researcher; Graduate Proposal Contributor.
- **NSF ECCS Collaborative Research: CPS: Medium; No. 2311087.**
Physics-Model-Based Neural Networks Redesign for CPS Learning and Control.
Support: \$197,000. June 2023 – May 2026
Role: Lead Graduate Researcher; Graduate Proposal Contributor.
- **NSF CNS Core: Small; No. 2140346.**
Enabling Real-time, Scalable and Secure Collaborative Intelligence on the Edge.
Support: \$499,861. Jan. 2022 – Dec. 2024
Role: Graduate Research Contributor; Graduate Proposal Contributor.

Industry Projects

- **[TIER IV: Design and Implementation of Infrastructure Programming Interface \(IPI\)](#)**.
Support: \$60,000. 2024 – 2025
Role: Lead Graduate Researcher and Project Lead; Graduate Proposal Contributor.
- **[Toyota: Open Intersection Reference Design and Implementation](#)**.
Support: \$60,000. 2023 – 2024
Role: Lead Graduate Researcher and Project Lead; Graduate Proposal Contributor.
- **[TIER IV: Investigating the Effect of C-V2X on Autoware.Universe](#)**.
Support: \$50,000. 2025 – 2026
Role: Lead Graduate Researcher and Project Lead; Graduate Proposal Contributor.

RESEARCH SYSTEMS & OPEN-SOURCE PLATFORMS

Runtime, shared-infrastructure, and sim-to-real systems across embedded computers and two full-scale vehicles.

- **[Autoware Foundation leadership](#)**. API Workgroup Lead; Technical Steering Committee and Strategic Planning Committee member. Defines APIs, middleware, V2X interoperability, and development priorities for the world's largest open-source autonomous-driving software stack, used by 500+ companies and deployed across 30 vehicle types in 20 countries.
- **[SIM/DAVOS/DaVoS](#)**. Designed newest-frame shared-memory sensor transport for ROS 2/Autoware. Reduced maximum observed latency by up to 98% and p99 by approximately 96%; the measured time saving equals 13.6 ft of travel at 40 mph, before accounting for downstream processing or braking. **Outputs:** [SIM](#); [DAVOS](#); [DaVoS](#); [code](#).
- **[IPI/InfraSense](#)**. Designed a CV/CAV API and authenticated roadway gateway; evaluated C-V2X/private 5G under mobility, load, weak signal, and restarts. **Outputs:** Edge4AV; [IPI](#); InfraSense.
- **[BlueICE/CARScenes/S-Score](#)**. Built tools for sim-to-real validation, scenario data, and dataset coverage. **Outputs:** [BlueICE](#); [CARScenes](#); [S-Score](#).

TEACHING & MENTORING EXPERIENCE

Designed the autonomous-driving course, taught two graduate courses, and led a hands-on program for high-school students.

Graduate-Level Courses

- **[Introduction to Autonomous Driving](#)**, University of Delaware, CIS. Fall 2025, Fall 2026
Role: Course Designer and Instructor. Designed and taught one of three designated courses in UD's autonomous-driving specialization for M.S. AI and CS students. The inaugural Fall 2025 class enrolled 20+ students across multiple departments; the current Fall 2026 offering enrolls 20 students. The course uses 1/10-scale and full-size autonomous vehicles. **Companion Websites:** [Demo](#); [Book Companion](#).
- **[Introduction to Mobile Robot Programming](#)**, University of Delaware, CIS. Summer 2024
Role: Instructor of Record. Taught project-based robot programming with ICAT and Autoware.

Pre-College Summer Program

- **[ADA for High-School Students](#)**, University of Delaware. Summer 2025, Summer 2026
Role: Program Designer and Instructor. Designed and taught the inaugural five-day Autonomous Driving Academy for High-School Students, serving 14 students from seven states. Students presented and demonstrated their autonomous-driving projects to parents and friends; the academy returned for a second offering in 2026.

Research Mentoring & Student Programs

- **CISC Graduate Promotion Unit (GPU).** 2024 – present
Role: Founder and President. Links undergraduates to researchers and graduate students to alumni.
- **Undergraduate Research Mentor.** 2024
Vertically Integrated Projects (VIP): Dustin (Duy Duc) Tran.
DSU–UD Summer Engineering Research Experience: Jared Bryant.

INVITED TALKS & TUTORIALS

Five invited research talks plus selected tutorials and live autonomous-vehicle demonstrations.

Invited Talks

- “A Closed-Loop System-Level Digital Twin to Evaluate Connected and Autonomous Mobility.”
Digital Twins for Intelligent Vehicles: From Agent-Level to System-Level Twins Workshop, IEEE Intelligent Vehicles Symposium. 2026
- “Digital Twins for Autonomy Verification and Validation.”
UCI CADRI Workshop, University of California, Irvine. 2025
- “Design of V2X Node in Autoware.”
Autoware Foundation API Workgroup. 2025
- “The Need for Digital Twins and Simulation for Autonomous Vehicle Testing.”
College of William & Mary. 2024
- “Using Digital Twin for AV System Verification.”
Autoware Foundation Centers of Excellence. 2024

Tutorials, Demonstrations & Selected Presentations

- “The Autoware Pipeline—Concepts and Basic Use.”
IEEE IV Autoware Tutorial. 2026
- “CARLab: CAV in the Autoware Ecosystem.”
IEEE IV Autoware Workshop. 2026
- “STARTaxi Robotaxi Demonstration.” [Demo Video 1](#); [Demo Video 2](#).
IEEE MOST Demonstration. 2025
- “An Advanced Framework for Ultra-Realistic Simulation and Digital Twinning for AV.”
Transportation Research Board Annual Meeting. 2025
- “Sim-to-Real and Real-to-Sim Indoor Testbed.”
IEEE MOST Tutorial. 2024
- “Autoware and CAV Tutorials.”
AutoSens Detroit and Detroit Auto Show. 2023, 2024

PROFESSIONAL SERVICE

Conference organization, program committees, and peer review.

- **IEEE MOST 2025.** Field Demo Chair.
- **SE4ADS at ICSE 2025.** Program Committee Member.
- **CAV@ASPLOS 2024.** Program Committee Member.
- **Reviewer, 2022–2026.** AAAI; CVPR; TRB Annual Meeting; IEEE WCNC, VTC, SEC, and MOST; ACM Computing Surveys; and Multimedia Tools and Applications.

INDUSTRY RESEARCH PROJECTS

Three automotive R&D appointments spanning AV perception, driver interaction, and cyber-physical testing, with production deliverables and an issued patent.

Torc Robotics: Autonomous-Vehicle Perception Data

- **Role:** Machine Learning Engineer L2, full-time co-op. Summer 2023
Project: Perception-model training data for autonomous vehicles.
Output: Production Perception-Training Datasets.

Joyson Safety Systems: Driver Monitoring and Steering-Wheel Interaction

- **Role:** Integration Engineer. Oct. 2019 – Oct. 2020
Project: GM Hands-On-Wheels driver-interaction program.
Output: GM Hands-On-Wheels Systems Integration and Validation.

UISEE Technology: Cyber-Physical Testing for Autonomous Shuttles

- **Role:** R&D Intern, Autonomous Driving Division. Summer 2018
Project: Cyber-physical test-system design for autonomous shuttles.
Output: Issued Chinese Patent [CN109357900B](#). Co-Inventor.